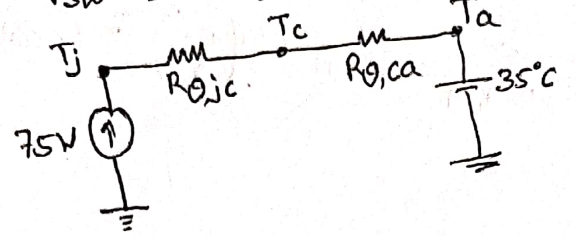
(5) $P_{cond} = 50W$

$P_{sw} = 10^{-3} f_s = 25W$




$$\frac{T_j - T_a}{(R_{\theta,jc} + R_{\theta,ca})} = 75W$$

$$\boxed{\left[\frac{T_j - T_a}{75}\right] - R_{\theta,jc} = R_{\theta,ca}}$$

$$\boxed{R_{\theta,ca} = \left[\frac{150-35}{75}\right] - 1 = 0.533\ °C/W.}$$

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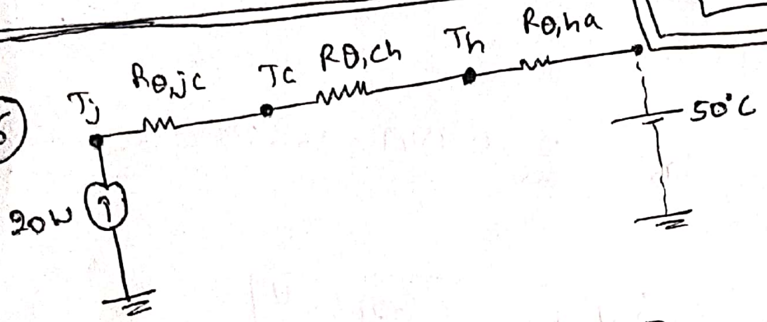
(6)




$$\frac{T_j - T_a}{20} = R_{\theta jc} + R_{\theta ch} + R_{\theta ha}$$

$$\frac{T_j - 50}{20} = 0.5 + 1.5 + R_{\theta,ha}$$

$$\therefore R_{\theta,ha} = \left[\frac{100-50}{20}\right] - 0.5 - 1.5$$

$$\boxed{R_{\theta,ha} = 0.5\ °C/W}$$

Now, $T_c = T_j - [R_{\theta,jc} \times 20W]$

$$\boxed{T_c = 100 - 10 = 90°C}$$

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(7) $$C = -\frac{(T_s/2)}{R \ln\left[\frac{V_{min}}{V_{max}}\right]} = -\frac{(T_s/2)}{\left[\frac{V_o^2}{P}\right] \ln\left(\frac{V_{min}}{V_{max}}\right)}$$

for (a) $$C = -\frac{20 \times 10^{-3}}{\left[\frac{\left[\frac{325.269+310.269}{2}\right]^2}{500}\right] \times \ln\left[\frac{310.269}{325.269}\right]} = 2.098\,mF$$

for (b) $$C = \frac{-10 \times 10^{-3}}{\left[\frac{\left[\frac{325.269+310.269}{2}\right]^2}{500}\right] \ln\left[\frac{310.269}{325.269}\right]} = 1.0488\,mF$$

for (c) $$C = -\frac{[(20 \times 10^{-3}) \div 6]}{\left[\frac{\left[\frac{565.685+550.685}{2}\right]^2}{500}\right] \ln\left[\frac{550.685}{565.685}\right]} = 199.046\,\mu F.$$

(8) output voltage of a q-pulse rectifier

$$V_{or} = V_m \frac{q}{\pi} \sin\left(\frac{\pi}{q}\right)\left[1 - \sum_{n=q,2q\ldots}^{\infty} \frac{2}{n^2-1} \cos\left(\frac{n\pi}{q}\right) \cos(n\omega t)\right]$$

for $q=6$,

$$V_{or} = 0.9549 V_m \left[1 + \frac{2}{35}\cos(6\omega t) - \frac{2}{143}\cos(12\omega t) + \cdots\right]$$

first dominant non-DC frequency component voltage $\quad V_{or,6} = \frac{2}{35} \cdot 0.9549 \times 565.685 = \underline{30.867V.}$

40dB attenuation by LC filter $\therefore \boxed{V_{out,6} = 0.30867V}$.

Load is 100KW $\quad \therefore R = \frac{V_o^2}{P} = \frac{(0.9549 \times 565.685)^2}{100 \times 10^3} = \underline{2.9179\Omega}$

$$\therefore \frac{1}{I_o} = \frac{0.9549 \times 565.685}{100 \times 10^3} \Rightarrow \boxed{I_o = 185.126A}$$

$$\boxed{\therefore \omega L = \frac{30.867 + 0.30867}{\left[\frac{0.2 \times 185.126}{2}\right]} = 1.68H.} \Rightarrow \boxed{L = 0.9mH}$$

now, $\frac{V_o(s)}{V_i(s)} = \frac{1/LC}{s^2 + \frac{1}{RC}s + \frac{1}{LC}} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$

$$20\log_{10}\left|\frac{V_o(s)}{V_i(s)}\right| = 20\log_{10}\left|\frac{1}{\left[1 - \frac{\omega^2}{\omega_n^2}\right] + j\,\frac{2\xi\omega}{\omega_n}}\right| = -40dB.$$

$$\frac{\omega^4}{\omega_n^4} - 2\frac{\omega^2}{\omega_n^2} + 4\xi^2\frac{\omega^2}{\omega_n^2} + 1 = 10,000$$

$$\boxed{\left(\frac{\omega^2}{\omega_n^2}\right)^2 + (4\xi^2 - 2)\left(\frac{\omega^2}{\omega_n^2}\right) - 9,999 = 0}$$

$$\frac{\omega^2}{\omega_n^2} = \frac{-(4\xi^2-2) \pm \sqrt{(4\xi^2-2)^2+39{,}996}}{2}$$

for $\xi = 0 \Rightarrow \frac{\omega^2}{\omega_n^2} = 101$

for $\xi = 0.5 \Rightarrow \frac{\omega^2}{\omega_n^2} = 100.49$

$$\boxed{\text{for } \xi = 0.707 \Rightarrow \frac{\omega^2}{\omega_n^2} = 100}$$

for $\xi = 1 \Rightarrow \frac{\omega^2}{\omega_n^2} = 99.$

$$\omega_n^2 = \frac{(2\times\pi\times300)^2}{100} \Rightarrow \boxed{\omega_n = 2\times\pi\times30}$$

Now, $\omega_n = \frac{1}{\sqrt{LC}} = 2\times\pi\times30.$

$$C = \frac{1}{(2\times\pi\times30)^2 \times L} = \frac{1}{(2\times\pi\times30)^2 \times \cancel{1.68}}$$

0.9mH.

$$\boxed{C = \cancel{16.75\,\mu F}} = 31.2\,mF.$$

9

(2) $P_{Load} = \frac{V_o^2}{R} \Rightarrow V_o = \sqrt{P_{load} \times R} = \sqrt{3900 \times 37}$

$\therefore \boxed{V_{out} = 379.8684V}$

(1) $V_{rms} I_{rms} = 3900W$

$I_{rms} = \frac{3900}{230} = 16.9565A$

$\therefore \boxed{I_{max} = 23.9801A}$

(3) $\boxed{I_{out} = \sqrt{\frac{P_{load}}{R}} = \sqrt{\frac{3900}{37}} = 10.2667A}$

10 Now, W.K.T the PFC shapes the inductor current like

$I_{in} = I_L = I_m |\sin \omega t|$

for boost converter, $\frac{I_{out}}{I_{in}} = (1-D) = \frac{I_{out}}{I_m |\sin \omega t|}$

Now, CCM

$$V_{out} = \frac{V_{in}}{1-D} = \frac{V_m |\sin \omega t|}{\frac{I_o \cancel{\quad}}{I_m (\sin \omega t)}}$$

$$V_{out} = \frac{V_m . I_m}{I_o} |\sin^2 \omega t|$$

$$\boxed{V_{out} = \frac{V_m I_m}{2 I_o} - \frac{V_m I_m}{2 I_o} |\cos(2\omega t)|}$$

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